On the number of certain Galois extensions of local fields

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Abstract: In this paper, we will calculate the number of Galois extensions of local fields with Galois group A_n and S_n.
Keywords: Local fields, Galois closure, ramified extension.
MSC: 11S15, 11S20.

1 Introduction

Let p be a prime, F a finite extension of p-adic field $\mathbb{Q}_p$ with $[F:\mathbb{Q}_p]=m$. Let k be its residue field with $[k:\mathbb{Z}/p\mathbb{Z}]=f$. It's clear $f|m$. Let π be a uniformizer of F and e be the absolute ramification index of F/Q_p. Hence $m=ef$. And let μ_l denote the set of l-th roots of unity. All notations are standard if not explained.

Since the number of the extensions of the local field with the given degree is finite, see [2]. One can ask for the formula that gives the number of extensions of a given degree. Krasner [1] gave such a formula, and Serre [5] also computed the number of extensions using a different method in the proof of his famous "mass formula". Paul and Roblot [6] gave another proof for the formula. Similarly one can also ask for the formula that gives the number of Galois extensions of the given degree. In particular, it is possible to ask for the formula that gives the number of the Galois extensions with prescribed finite Galois group G. We denote the number by $\nu(F,G)$. If G is a p-group with $\mu_p \nsubseteq F$, Šafarevič [3] gave an explicit formula for the number of the G-extensions of F:

$$\nu(F,G)=\frac{1}{|Aut(G)|}(\frac{|G|}{p^d})^{m+1}\prod_{i=1}^{d-1}(p^{m+1}-p^i),$$

where d is the minimal number of generators of G. If G is a p-group, and $\mu_p \subset F$, Yamagishi [7] obtained a formula for $\nu(F,G)$.

In this paper, we will calculate the number of S_n-extensions and A_n-extensions of F, where S_n is n-th symmetric group an A_n is n-th alternating group, that is,

Theorem 1.1 (1) *Suppose the prime $p\neq 3$, then*

$$\nu(F,S_3)=\begin{cases}0 & \text{if } \mu_3\subset F,\\ 1 & \text{if } \mu_3\nsubseteq F.\end{cases}$$

(2) *Suppose* $p = 3$, *then*

$$\nu(F, S_3) = \begin{cases} 3^{m+1} - 3 & \text{if } \mu_3 \subset F, \\ 3^m + \frac{3^{m+1}}{2} - \frac{3}{2} & \text{if } \mu_3 \not\subset F. \end{cases}$$

Theorem 1.2 (1) *Suppose the prime* $p \geqslant 3$, *then*

$$\nu(F, S_4) = \nu(F, A_4) = 0.$$

(2) *Suppose* $p = 2$, *then*

$$\nu(F, A_4) = \begin{cases} 4(2^{2m} - 1)/3 & \text{if } \mu_3 \subset F, \\ (2^{2m} - 1)/3 & \text{if } \mu_3 \not\subset F; \end{cases}$$

$$\nu(F, S_4) = \begin{cases} 0 & \text{if } \mu_3 \subset F, \\ 2^{2m+1} - 1 & \text{if } \mu_4 \not\subseteq F \text{ and } m \text{ is even and } f = 1, \\ 2^{2m} - 1 & \text{otherwise.} \end{cases}$$

2 Some lemmas

Lemma 1 *Suppose* $n \geqslant 5$, *there does not exist a Galois extension of* F *with Galois group* S_n *or* A_n.

Proof: If F is a local field, then the Galois group $\mathrm{Gal}(K/F)$ is a solvable group for every finite Galois extension K of F. It is well-known that A_n and S_n is not a solvable group for $n \geqslant 5$.□

It is well-known that the number of S_2-extensions and A_3-extensions of local field by local classfield throry. So we only need to calculate the number of Galois extensions of field F with Galois group S_3 ,S_4 and A_4. The following lemma plays important role in our calculation.

Lemma 2 *Let* K *be a Galois extension of* F *with the Galois group* G. *For any subgroup* A *of* G, *let* F_A *be the field fixed by* A, *then the Galois closure* $cl(F_A)$ *of* F_A *is a subfield of* K *and* $Gal(K/cl(F_A)) = N_G(A)$, *where* $N_G(A) = \{s \in A | t^{-1}st \in A, \text{for all } t \in G\}$.

We can get some Galois extensions from some non-Galois extensions by its' Galois closure. For example, if G=S_3, D_8, A_4 and S_4, we can choose A to be order 2, order 2, order 3 and order 6 non-normal subgroup respectively of G, where D_8 is the 2-sylow subgroup of S_4. We know A is isomorphic to $\mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/3\mathbb{Z}$ and S_3 respectively. And their normalization is trivial. By the above lemma, the Galois extensions of F with Galois group S_3 can be gotten by the Galois closure of extension of degree 3 of F, and the Galois extensions of F with Galois group D_8, A_4 and S_4 can be gotten by the Galois closure of extensions of degree 4 of F.

Let $M(n)$ be the set of all extensions of degree n of F. Let $Ab(n)$ be the set of abelian extensions of degree n of F. And let $M(G)$ be the set of Galois extensions of F with the Galois group G. Let K be the Galois closure of an extension of degree n of F . The Galois group $\mathrm{Gal}(K/F)$ is a subgroup of S_n. Obviously the order of $Gal(K/L)$ must be divided by n. So there are the following two maps:

$$f: \quad M(3) \to Ab(3) \cup M(S_3)$$

and

$$g: \quad M(4) \to Ab(4) \cup M(D_8) \cup M(A_4) \cup M(S_4)$$

by

$$L \to cl(L).$$

The two maps are surjective. Any inverse image L of an element K of $M(G)$ is a subfield of K and L is not a Galois extension of F if G is not an abelian group. By Galois theory, the Galois group $\mathrm{Gal}(L/K)$ is not a normal subgroup of G. For $G = S_3, D_8, A_4$ and S_4, we consider respectively the number of the non-normal subgroup isomorphic to $\mathbb{Z}/2\mathbb{Z}$,$\mathbb{Z}/2\mathbb{Z}$,$\mathbb{Z}/3\mathbb{Z}$ and S_3. The subgroups are non-normal except there is an order-2 normal subgroup in D_8. So the number of the inverse image of any element of $M(S_3), M(D_8), M(A_4)$ and $M(S_4)$ are $3, 4, 4, 4$ respectively. Let $|S|$ denote cardinality of a finite set S. Let $\nu(F, G)$ denote the number of $M(G)$. So there is the following result.

Lemma 3

$$|M(3)| = |Ab(3)| + 3\nu(F, S_3),$$
$$|M(4)| = |Ab(4)| + 4\nu(F, D_8) + 4\nu(F, A_4) + 4\nu(F, S_4).$$

3 The proof of theorem 1.1

In the following, we denote $m = [F : \mathbb{Q}_p]$ and e is the absolute ramification index of F and $q = p^f$ is the number of elements of the residue field of F.
Proof: (1) For $p \neq 3$,

$$M(3) = \{K|\ [K : F] = 3, K \text{ is a tamely ramified extension of } F\}.$$

(i) If 3-th roots of unity are contained in F, then $M(3) = Ab(3)$. By lemma 3

$$\nu(F, S_3) = 0.$$

(ii)If 3-th roots of unity are not contained in F, then $|M(3)| = 4$, $|Ab(3)| = 1$. By lemma 3

$$\nu(F, S_3) = 1.$$

(2) For $p = 3$, by Krasner'theorem [3],

$$|M(3)| = 3q^e + 6(q-1)(\sum_{a=0}^{e-1} q^a) + 1 = 9q^e - 5.$$

Suppose $\mu_3 \not\subset F$, then

$$|Ab(3)| = \frac{1}{2}(\frac{3}{3})^{m+1}(3^{m+1} - 1) = \frac{3^{m+1} - 1}{2} = \frac{3q^e - 1}{2}.$$

Suppose $\mu_3 \subset F$, then

$$|Ab(3)| = 4.$$

By lemma 3

$$\nu(F, S_3) = \begin{cases} \frac{5q^e - 3}{2} & \text{if } \mu_3 \not\subset F, \\ 3q^e - 3 & \text{if } \mu_3 \subset F. \end{cases}$$

□

4 The proof of theorem 1.2

First we give some propositions.

Proposition 4.1 *If the prime $p \geqslant 3$, then*

$$\nu(F, S_4) = \nu(F, A_4) = 0.$$

Proof: Suppose K is a Galois extension of F with Galois group S_4. There must exist intermediate fields F^{tr} and F^{ur} such that $\mathrm{Gal}(K/F^{tr})$ is a p-group, $\mathrm{Gal}(F^{tr}/F^{ur})$ and $\mathrm{Gal}(F^{ur}/F)$ are cyclic groups. By Galois theory, there is a p-group S' which is a normal subgroup of S_4. Since $p \geqslant 3$, S' must be (1). Since S_4 has not a cyclic normal subgroup S such that S_4/S is also cyclic, a contrary is derived. The result follows similarly for A_4.□

Proposition 4.2 *If $p = 2$,*

$$\nu(F, A_4) = \begin{cases} 4(2^{2m} - 1)/3 & \text{if } \mu_3 \subset F, \\ (2^{2m} - 1)/3 & \text{if } \mu_3 \not\subset F. \end{cases}$$

Proof: Let K be an A_4-extension of F. Since K_4 is a normal subgroup of A_4, there exists a (unique) Galois subfield F' of degree 3 of K, where $K_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. By [2],

$$|F'^*/(F'^*)^2| = 4q^{3e},$$

$$|F^*/(F^*)^2| = 4q^e.$$

It is clear that the natural map of $F^*/(F^*)^2 \to F'^*/(F'^*)^2$ is an injection. We consider the action on $F'^*/(F'^*)^2$ of the Galois group $\mathrm{Gal}(F'/F)$.

(1) Denote $G' = Gal(F'/F)$. Then the following result holds.

$$(F'^*/(F'^*)^2)^{G'} = F^*/(F^*)^2.$$

In the following we will prove it. Let $a \in F'^*/(F'^*)^2$ be a fixed point of $\mathrm{Gal}(F'/F)$ and $a \neq 1$ in $F'^*/(F'^*)^2$. Then $F'(\sqrt{a'})$ is a Galois extension of F, where a' represents a lifting of a in F'^*. There isn't an order-2 normal subgroup in S_3, so

$$\mathrm{Gal}(F'(\sqrt{a'})/F) = \mathbb{Z}/6\mathbb{Z}.$$

Let F'' be the fixed field of the normal subgroup $\mathbb{Z}/3\mathbb{Z}$. There exists an element $b \in F^*/(F^*)^2$, such that

$$F'' = F(\sqrt{b}).$$

Then

$$F'(\sqrt{a'}) = F'(\sqrt{b}).$$

So $a = b$ in $F'^*/(F'^*)^2$.

(2) Let σ be a generator of G'. If $x \notin F^*/(F^*)^2$. Then there are the following two case:
(i) $N_{F'/F}(x) = 1$ in $F^*/(F^*)^2$,

(ii) $N_{F'/F}(x) \neq 1$ in $F^*/(F^*)^2$.
In (i), the field $F'(\sqrt{x}, \sqrt{\sigma x})$ is an A_4-extension of F since the Galois group of $F'(\sqrt{x}, \sqrt{\sigma x})/F$ isomorphic to $K_4 \rtimes \mathbb{Z}/3\mathbb{Z}$, where $K_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.
In (ii),

$$F'(\sqrt{x}, \sqrt{\sigma x}, \sqrt{\sigma^2 x}) = F'(\sqrt{x/\sigma(x)}, \sqrt{\sigma x/\sigma^2 x}, \sqrt{N_{F'/F}(x)}),$$

So $F'(\sqrt{x}, \sqrt{\sigma x}, \sqrt{\sigma^2 x})$ is a $(A_4 \times \mathbb{Z}/2\mathbb{Z})$-extension of F. And for any $(A_4 \times \mathbb{Z}/2\mathbb{Z})$-extension K, it is generated by an A_4-extension of F and an extension of degree 2 of F. Denote the unique extension of degree 3 by F', then there exist $x \in F'^*/(F'^*)^2$ and $a \in F^*/(F^*)^2$ satisfying $x \notin F^*/(F^*)^2$, $N_{F'/F}(x) \in (F^*)^2$ and $a \notin (F^*)^2$, such that $K = F'(\sqrt{x}, \sqrt{\sigma x}, \sqrt{a})$. It is easy to see

$$K = F'(\sqrt{ax/\sigma x}, \sqrt{a\sigma x/\sigma^2 x}, \sqrt{a\sigma^2 x/x})$$

since $N_{F'/F}(x) = x\sigma x\sigma^2 x = 1$ in $F^*/(F^*)^2$, then $\sigma x/\sigma^2 x = x\sigma^2 x/\sigma^2 x = x$ in $F'^*/(F'^*)^2$.
Consider K as an extension of F', there exist 7 subfields with order 2 over F' which are one-to-one correspondent to $\{y, \sigma y, \sigma^2 y, y\sigma y, \sigma y\sigma^2 y, y\sigma^2 y, N_{F'/F}(y)\}$, where $y = ax/\sigma x$. And $N_{F'/F}(y) = a \neq 1$ in $F^*/(F^*)^2$. The $\mathrm{Gal}(F'/F)$-orbits are $\{y, \sigma y, \sigma^2 y\}$, $\{y\sigma y, \sigma y\sigma^2 y, y\sigma^2 y\}$, and $\{N_{F'/F}(y)\}$. So any $(A_4 \times \mathbb{Z}/2\mathbb{Z})$-extension of F can be generated by the element of (ii) and the claim follows.

$$\nu(F, A_4 \times \mathbb{Z}/2\mathbb{Z}) = \nu(F, A_4)\nu(F, \mathbb{Z}/2\mathbb{Z}).$$

(i) Suppose $\mu_3 \not\subset F$, F' is the unique unramified extension of degree 3 of F, so

$$3\nu(F, A_4) + 3\nu(F, A_4 \times \mathbb{Z}/2\mathbb{Z}) = 4q^{3e} - 4q^e,$$

$$\nu(F, A_4 \times \mathbb{Z}/2\mathbb{Z}) = \nu(F, A_4)(4q^e - 1).$$

Then

$$\nu(F, A_4) = (q^{2e} - 1)/3.$$

(ii) Suppose $\mu_3 \subset F$, F' is the unique unramified extension and 3 totally ramified extensions of degree 3 of F, so

$$3\nu(F, A_4) + 3\nu(F, A_4 \times \mathbb{Z}/2\mathbb{Z}) = 4(4q^{3e} - 4q^e),$$

$$\nu(F, A_4 \times \mathbb{Z}/2\mathbb{Z}) = \nu(F, A_4)(4q^e - 1).$$

Then

$$\nu(F, A_4) = 4(q^{2e} - 1)/3.$$

□

Proposition 4.3 *Let $p = 2$,*

$$\nu(F, S_4) = \begin{cases} 0 & \text{if } \mu_3 \subset F, \\ 2^{2m+1} - 1 & \text{if } \mu_4 \not\subseteq F \text{ and } m \text{ is even and } f = 1, \\ 2^{2m} - 1 & \text{otherwise.} \end{cases}$$

Proof: By Krasner's theorem [3],

$$|M(4)| = 16q^{3e} - 4q^{2e} - 5.$$

By the local classfield throry and the dual theory of the finite abelian group, the following equation holds:

$$|Ab(4)| = |\{S : \ S \text{ is the subgroup of order 4 of } F^*/(F^*)^4\}|.$$

Let T_1 be the set consisting of the elements of order $\leqslant 2$ in $F^*/(F^*)^4$, T_2 be the set consisting of the elements of order 4 in $F^*/(F^*)^4$. The sequence

$$0 \to T_1 \to F^*/(F^*)^4 \to (F^*)^2/(F^*)^4 \to 0$$

is exact. The third map is $a \mapsto a^2$. So

$$|T_1| = |F^*/(F^*)^4|/|(F^*)^2/(F^*)^4|.$$

Suppose $\mu_4 \nsubseteq F$, then

$$|T_1| = 4q^e,$$
$$|T_2| = 8q^{2e} - 4q^e.$$

Then

$$|Ab(4)| = |T_2|/2 + (|T_1| - 1)(|T_1| - 2)/6 = 20q^{2e}/3 - 4q^e + 1/3.$$

Suppose $\mu_4 \subset F$, then

$$|T_1| = 4q^e,$$
$$|T_2| = 16q^{2e} - 4q^e.$$

Then

$$|Ab(4)| = |T_2|/2 + (|T_1| - 1)(|T_1| - 2)/6 = 32q^{2e}/3 - 4q^e + 1/3.$$

Since D_8 is a 2-group, by theorem 2.2 [7],

$$\nu(F, D_8) = \begin{cases} q^e(q^e - 1)(4q^e - 1) & \text{if } \mu_4 \subset F \text{ or } \mu_4 \nsubseteq F \text{ and } m \text{ is even and } f = 1, \\ q^e(2q^e - 1)^2 & \text{otherwise.} \end{cases}$$

By lemma 3,
(i) If $\mu_4 \subset F$, then

$$\nu(F, S_4) = (|M(4)| - |Ab(4)|)/4 - \nu(F, A_4) - \nu(F, D_8) = 4(q^{2e} - 1)/3 - \nu(F, A_4).$$

(ii) If $\mu_4 \nsubseteq F$, and m is odd or m is even and $f \geqslant 2$, then

$$\nu(F, S_4) = (|M(4)| - |Ab(4)|)/4 - \nu(F, A_4) - \nu(F, D_8) = 4(q^{2e} - 1)/3 - \nu(F, A_4).$$

(iii) If $\mu_4 \nsubseteq F$ and m is even and $f = 1$, then

$$\nu(F, S_4) = (|M(4)| - |Ab(4)|)/4 - \nu(F, A_4) - \nu(F, D_8) = (7q^{2e} - 4)/3 - \nu(F, A_4).$$

By proposition 4.2, so

$$\nu(F, S_4) = \begin{cases} 0 & \text{if } \mu_3 \subset F, \\ 2q^{2e} - 1 & \text{if } \mu_4 \not\subseteq F \text{ and } n \text{ is even and } f = 1, \\ q^{2e} - 1 & \text{otherwise.} \end{cases}$$

□

Remark: Since K_4 is a normal subgroup of S_4 and $S_4/K_4 \cong S_3$, there exists an S_3-subextension in an S_4-extension of F by Galois theory. If $\mu_3 \subset F$ and $p \neq 3$, then $\nu(F, S_3) = 0$. So $\nu(F, S_4) = 0$. This gives another proof for a case of $\nu(F, S_4)$.

Using these propositions, the proof of theorem 1.2 is obtained. This completes the proof of theorem 1.2.

5 Examples

Example 5.1 *Let $F = \mathbb{Q}_p$, then*
(1)

$$\nu(\mathbb{Q}_p, S_3) = \begin{cases} 6 & \textit{if } p = 3, \\ 0 & \textit{if } p \equiv 1 \mod 3, \\ 1 & \textit{if } p \equiv 2 \mod 3. \end{cases}$$

(2)

$$\nu(\mathbb{Q}_p, A_4) = \begin{cases} 1 & \textit{if } p = 2, \\ 0 & \textit{if } p > 2. \end{cases}$$

(3)

$$\nu(\mathbb{Q}_p, S_4) = \begin{cases} 3 & \textit{if } p = 2, \\ 0 & \textit{if } p > 2. \end{cases}$$

(4)

$$\nu(\mathbb{Q}_p, S_n) = \nu(\mathbb{Q}_p, A_n) = 0 \quad (n \geqslant 5).$$

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